INFO 6150 Web Design and User Experience Engineering: Spring 2026: Final project

For your final project, you will create a website and/or application individually. You will create a simple prototype for an idea you have in mind (as long as there are APIs for it).

There will be three phases to the project:

1. **Propose a project** and submit the proposal to me for assessment. I may suggest increasing or reducing the project's scope to make it manageable. You will have the chance to rectify your proposal. Afterwards, the project contents and scope are established. **Due date: March 13, 20256**
2. **Create the website or application**, along with all required documentation, as agreed with the instructor. The project needs to run (even if not perfectly) and, ideally, be deployed on the public web; at the very least, the code must be in a GitHub repo you own. **Due date: April 24, 2026,** at 4:30 pm Pacific Time.
3. **Presentation:** On **April 24, 2026**, everybody will have at most 12 minutes to **present their creations to the class**. Afterwards, the class will have up to 8 minutes to **respond**, i.e., ask relevant questions about the project's reasoning, the methods used, the project's future, etc. **Questions here may count as extra points on the final grade for whoever asks, up to 2 Canvas points per question.** If no questions are asked, the instructor may ask instead. The presenting person will also be evaluated on the quality of their answers (ie, whether they are well justified).

Grading

The final project accounts for 35% of the course grade.

The project has a maximum score of 350 points, distributed as follows:

- Requirements (documents and code): 200 points
- Final presentation: 120 points
- Answers to questions after presentation: 30 points

Requirements (200 points)

Projects must contain:

- Everything from the “base requirements” list, and
- Several “specialty requirements” described below.

Base requirements (150 points)

All projects **must have** at least:

- A short description of the project: why are you creating it, who are the users, and what is the expected result? **(5 points)**
- A persona analysis with at least two personas. **(20 points, 10 per persona)**
- Two user journeys, with at least two opportunities discovered on each user journey. **(20 points, 10 per user journey)**
- A complete sitemap, including all pages in the website or app. **(20 points)**
    - It must be clear what information goes on each page.
    - If there are processes, they must be explained with flowcharts, box diagrams, or through written algorithms.
- Low-resolution mockups for each page that is different on the site or app. They must be at least two. **(20 points, 10 per mockup)**
- A description of the typography and color palette, rooted in design principles and considering the users and use cases of the website or app. **(10 points)**
- The website or app is shown as code **(20 points)** and working on a window browser **(30 points)**.
    - It must exist in a GitHub repository owned by one of the teammates.
        - No other accounts, except those owned by the teammates, must have collaborated on the repository.
    - There must be at least two different pages or interfaces in the project.
    - HTML, CSS, and JavaScript must be of near-professional quality, as per the tests done on them.
        - Points may be discounted due to errors in coding style, inconsistent coding (such as erroneous HTML or CSS markup), JavaScript bugs, etc.
    - Functionality should work well.
        - If something does not work on time, this must be disclosed by the team.
        - Points may be discounted for errors or issues (for example, the website is not working at all, files are missing or not properly linked, basic functionality does not work well, etc.)
            - The basic idea here is that the team will commit to what they expect will work, and then deliver on it, mostly free of problems.
- A short description of the future of the project (if it will continue) and the team's learnings. **(5 points)**

Specialty requirements

Mix and match from the following requirements. **Your selection must reach at least 50 points.** The maximum number of points possible for specialty requirements is 50. Extra points cannot be used to supplement the "base requirements" grading.

Option 1: User research (25 points)

Before creating personas, learn from real people how they would use your product.

- Create a questionnaire and have 8 different people answer it. (This can be printed on paper, or it can be online using Google Forms or similar.) These must be people who would use your website or app.
- You should ask, at least:
    - How do they do, right now, whatever your website or app will do for them,
    - What problems do they have with the current way of doing things,
    - What ideas do they have for your product?
    - Any information that will help you categorize them: age range, education, etc. This information must be optional: if people do not feel comfortable answering, they should not answer.
- Then, **you must base your personas on the data you got from these questionnaires.**
    - Instead of imagining or inventing your personas, try to figure out patterns from the information you received. Who are these people? How to group them into one or two personas that make sense?

Extra requirements for this option:

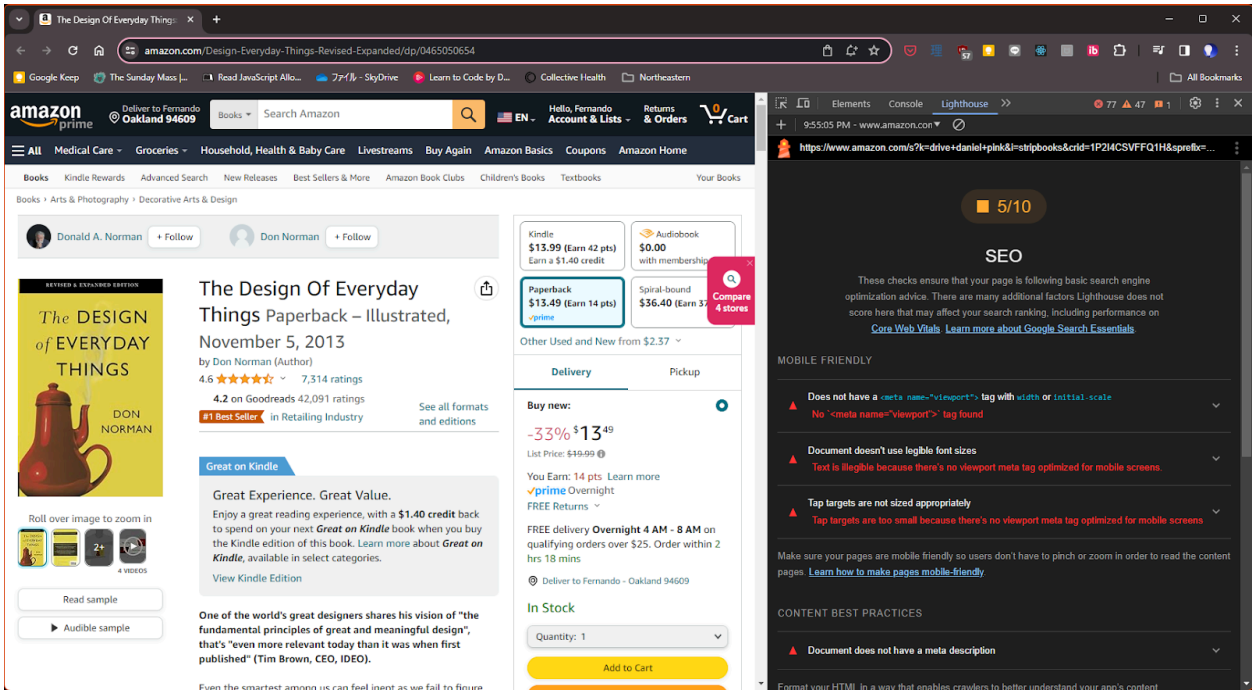
- You must present the questionnaires and the data received from them. (2 points per questionnaire, up to 16)
- You must explain, in a document, how the personas are related to the data in the questionnaires. (9 points)

Option 2: SEO master (10 points)

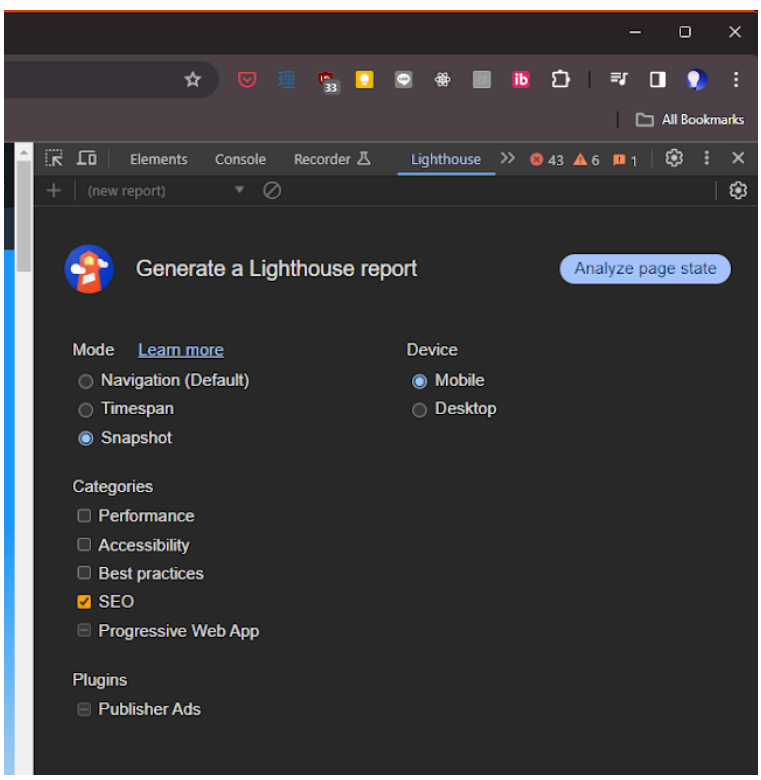
Your website or app must pass the Lighthouse SEO evaluation.

Extra requirements for this option:

- Pass the Lighthouse evaluation (1 point per SEO check). For example, this would get 5 points:



To get this report, you must use the SEO option on the Lighthouse panel of Chrome Dev Tools:



Notes:

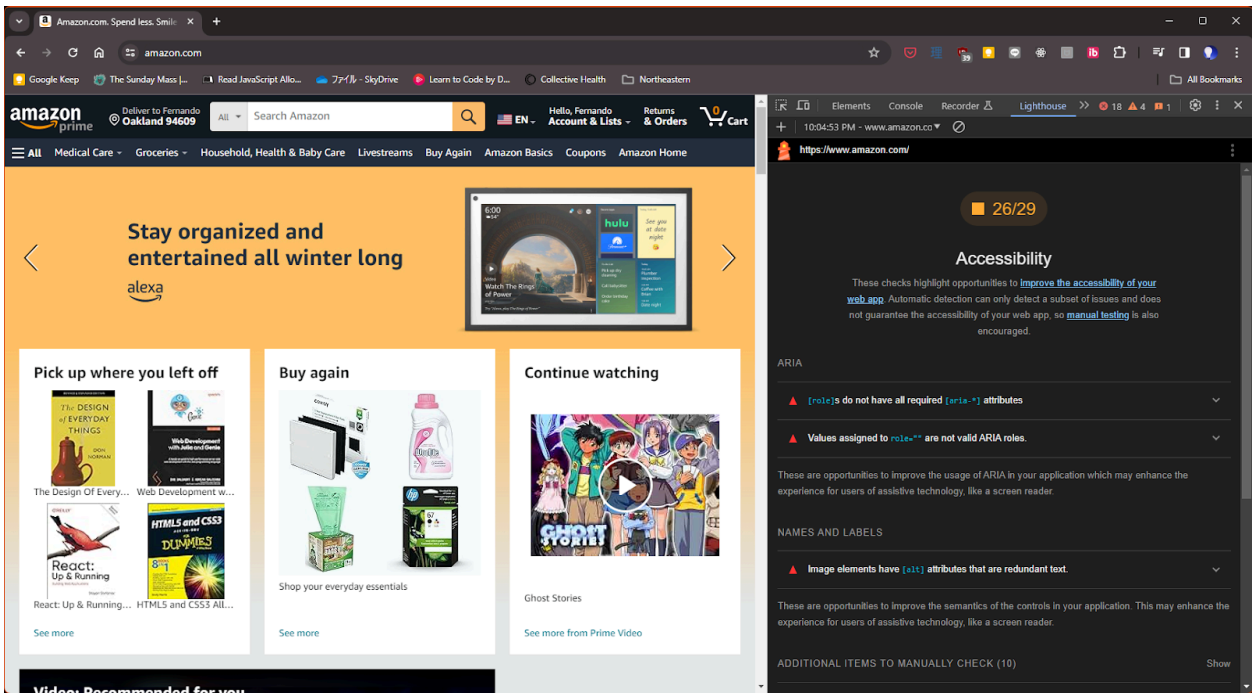
- Different websites might get a different “target”, depending on their content. In this example, it was 10. If you get to the target, you get the 10 Canvas points no matter what. Otherwise, you only get 1 Canvas point per check.
- The maximum is 10 Canvas points. You do not get more than 10 points if your website passes more than 10 SEO checks.

Option 3: Accessibility master (15 points)

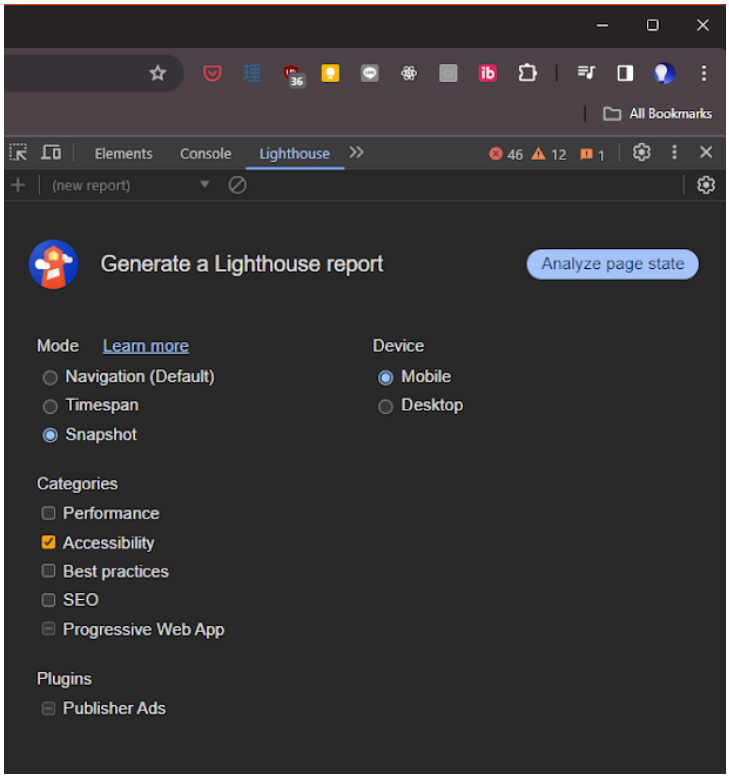
Your website or app must pass the Lighthouse accessibility evaluation.

Extra requirements for this option:

- Pass the Lighthouse evaluation (0.5 points per accessibility check). For example, this would get 0.5 + (26 / 2) = 13.5 points:



To get this report, you must use the accessibility option on the Lighthouse panel of Chrome Dev Tools:


All Bookmarks
Elements
Console
Lighthouse
46
12
1
(new report)
Generate a Lighthouse report
Analyze page state
Mode
Learn more
Navigation (Default)
Timespan
Snapshot
Device
Mobile
Desktop
Categories
Performance
Accessibility
Best practices
SEO
Progressive Web App
Plugins
Publisher Ads

Notes:

- Different websites might get a different “target”, depending on their content. In this example, it was 29. If you get to the target, you get the 15 Canvas points no matter what. Otherwise, you only get 0.5 Canvas points per check.
- The maximum is 15 Canvas points. You do not get more than 15 Canvas points if your website passes more than 30 accessibility checks.

Option 4: CSS master (15 points)
The website or app must demonstrate at least three CSS techniques not covered in the slides. These must be used judiciously: they must have a reason to be on the website, serving the objective of the design, and not just be inserted randomly on the page.
Extra requirements:
- Three techniques (5 points each).
    - Points may be discounted if the technique is not used properly or if it has no reason for being within the design. (For example, it collides with the design or distracts attention from the main objective of the page.)

Option 5: Delightful design (10 points)
The website or app must have extremely good usage of color, typography, and all design principles to create a professional-level presentation. This must apply to the general theme of the website.
Extra requirements:
- The design must be shown in both the low-resolution mockups (2 points) and executed on the website (5 points).
- The design flow must be effective: the user must be drawn to the most important part of the page first, the 2nd most important after that, and so on. (10 points: evaluated on two different pages, 5 points per page).
- There must be a written explanation for how the different design principles were used. This should also be explained orally during the presentation. (5 points)

Option 5: Effective design (25 points)
The design of the website or app must allow users to quickly do what they intend to do there. In addition, it must have extremely good usage of color, typography, and all design principles to create a professional-level presentation. All of the pages must capture the attention of the user, moving it towards the most important parts of the page.

You must do actual research with at least 2 potential users that represent your personas. For this:
- Reserve some time in the XR Lab (email the manager, Eric Lum, at e.lum@northeastern.edu for more information) and bring your users.
- Show your users either **a high-resolution mockup** or **a finished page**. Ask them to achieve a certain task there.
    - For example: buying a book on a library website, or sharing a photo on a social media app.

- Use the eye-tracking software available in the lab to see how they react.
- Get evidence that the user flow is effective, or that it needs improvement. If improvement is needed (eg, the users did not behave as you wanted them to), change the design accordingly.
    - If you change the design, you **do not** need to test it again - this would be difficult due to time constraints.
    - Evidence can be either printed or just shown as movies within your presentation. In any case, you need to explain them.
- Ask Eric for assistance on how to run the experiment and how to get the print-outs.

Extra requirements:
- The design must be shown in both the low-resolution mockups (2 points) and executed on the website (5 points).
- There must be a written explanation for how the different design principles were used. This should also be explained orally during the presentation. (5 points)
- Declare what the goal was for the users to achieve during your research. This must be directly related to the personas and their user journeys and to the goal of your website or app. (2 points)
- Present the evidence for the two users, showing how their eyes moved through the page as they tried to experiment. (6 points, 2 per user)
- Present a conclusion for your research: either that you demonstrated that your design was effective, or that you found it needed improvement. (5 points)
    - If improvement was needed, you **need** to implement those improvements in your final presentation.

Option 6: JavaScript master (15 points)
The website or app must demonstrate three JavaScript or React techniques not seen in class and must do so without any bugs that the instructor can find. Examples include usage of localStorage, cookies, custom React hooks, external libraries not seen in class, etc. The techniques must make sense within the app or website: they must aid in the website's ultimate goal and are not just randomly inserted into a page.

Extra requirements:
- The website must demonstrate each technique (5 points per technique).
    - A document must specifically mention each demonstrated technique.
    - Points may be discounted for any bug found.

Option 7: API integration (25 points)
The website or app uses a free web API to achieve some functionality in a mostly error-free way.

If choosing this option, you **must** consult with the instructor to carefully delimit the project. The project must use at least two "business objects". A business object can also be thought of as an

“entity” in a relational database. (For example, in a library system, this could be “books” and “users”).

Extra requirements:

- Two sets of functionality (use cases), as defined in conjunction with the instructor (10 points each).
    - For example: “leasing a book”, “returning a book”.
    - Points may be subtracted for bugs found. How strict this will be will go as per the difficulty of the use case: easier ones may be more strictly reviewed.
- A document explaining (5 points):
    - What API was used and why?
    - What other APIs existed but were not chosen?
    - What endpoints are being used, and what is their function?

Final presentation (120 points)

During the presentation, you will have a **strict limit** of 12 minutes. The presentation should explain:

- What the project is about,
- Why was it chosen?
- Each one of the requirements above (some may be just mentioned, but there must be some evidence, for example, shown on slides),
- Which specialty requirements were chosen, plus evidence for those extra requirements (again, this can be just shown on slides if the time is short).

Then, you must show the website or app at work, in a browser. This should take around 6 of the 12 minutes.

- Show whatever parts of the project are working and mention what was planned but not completed.
    - Whatever was not completed will not be evaluated for bugs.

Lastly, in one minute or so, you should mention what the future of the project and/or what lessons were learned are.

The presentation will be evaluated as follows:

- The presentation starts on time (15 points).
- The presentation ends on time (15 points).
- All of the requirements are explained (30 points).
    - Discounts may be given for missing requirements.
- The **completed** parts of the project work perfectly (50 points).
    - This means that during the project presentation, no bugs appear. Each bug may cause up to 5 points to be discounted. Points will be discounted on a sliding scale:
        - The more severe the bug is, the more points are discounted.

        - A very severe bug is “a page’s style does not load”, “a link is broken”, “a form will not send data”, badly executed CSS layouts (misaligned), etc.
        - A less severe bug is a typo, a header or paragraph style not showing correctly, a validation error in an input that does not completely break functionality, etc.
    - The less complex a feature is, the more points may be discounted.

Recommendations:
- Practice the presentation beforehand to make sure it starts and ends on time.
- Prepare slides to guide the presentation.
- Preload your slides and project them to the computer you will use to present.

Answers to questions after presentation (30 points)

Immediately after the presentation, you will have at most 8 minutes to answer questions from the instructor or other class members.

The quality of each answer will determine the points to be received. **Each answer can get a maximum of 15 points.** This means that you must receive at least two questions for a chance to get the full 30 points.

If you can answer more than two questions, each answer will be evaluated separately, and the two questions with the most points will be summed to get the final score.

The quality of the answers will be evaluated by the following rubric:
- The answer makes sense based on what is shown in the project and the requirements’ evidence. (5 points)
- The answer makes sense as per the concepts seen in class or researched independently; if challenged on this, you can provide a reasonable explanation. (10 points)

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Extra Canvas points for good questions

This can help students who need to recover points from missed or late assignments, or who are aiming for an excellent score. **This is your last chance to get extra points applied to your final score.**

While it’s not possible to come up with a recipe for what a good question is, there are some “ingredients”:

- It’s not a generic question; it’s directly related to what is seen in the project.
- It’s possibly related to the project's users and its objective.
- It’s not a repeat question, that is, something that someone else already asked (for this or another team’s project).
- It’s something that challenges the team to come up with a valid answer.

Each question will earn between 0 and 2 Canvas points for whoever asked it. Fewer points may be assigned depending on the quality of the “ingredients” above.

Questions must be asked respectfully and collaboratively. Question the project, the product, the process, and the techniques, not the people.